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Subject-Independent Model and Individual Calibration Algorithms for Determining Driver Drowsiness Based on Facial Landmarks

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***Abstract*:** *Vision-based driver drowsiness detection is often reported at above 95 % accuracy, yet such systems rarely reach production at that level of reliability. This paper argues that much of the gap comes from how the data are split. We extract eye aspect ratio, mouth aspect ratio and head-pose features from MediaPipe Face Mesh landmarks on the NTHU-DDD corpus, and train CNN, LSTM and GRU classifiers on five-second windows. Each model is evaluated twice: once with a random window split, and once with a subject-independent split in which no driver appears in both training and test data. The GRU reaches 97.8 % accuracy under the random split but only 88.4 % under the subject-independent one, a drop of 9.4 percentage points; the other two models behave the same way. Per-driver accuracy under the strict protocol ranges from 74.5 % to 96.1 %, so the mean figure hides drivers for whom the system is barely usable. We trace this spread to the anatomy of the eye: the alert-state EAR baseline varies from 0.198 to 0.305 across test drivers, so a single global threshold cannot serve all of them. Normalizing each feature against a short per-driver baseline recovers 3.9 points of accuracy and narrows the per-driver range to 84.2-96.8 %. We conclude that subject-independent evaluation and per-driver calibration should be standard practice in this field.*

Keywords: *driver drowsiness detection, facial landmarks, MediaPipe Face Mesh, eye aspect ratio, subject-independent evaluation, data leakage, recurrent neural networks, personalization..*

Introduction

Driver fatigue is a recognized cause of road deaths, and unlike most crash factors it gives warning before it becomes dangerous. Eyelids close for longer, blinks slow down, yawns appear, and the head

begins to nod. A camera facing the driver can see all of these signs, which is why vision-based drowsiness detection has become a standard component of driver monitoring systems.

The technical pipeline is now well established. A face detector locates the driver, a landmark model returns key points on the eyes and mouth, and simple geometric ratios describe how open each is. The eye aspect ratio (EAR) and mouth aspect ratio (MAR) are the two most common features. Early systems compared these ratios against fixed thresholds; later work fed sequences of them into recurrent networks, which can recognize the pattern of a slow blink or a yawn rather than a single frame.

Published accuracies are high. Figures above 95 % are routine, and several papers report more than 98 %. Deployment experience does not match this. One reason is well documented: many studies split their data by frame or by clip, not by driver, so frames from the same person and the same recording session appear in both the training and the test set. A model can then score well by recognizing the driver, the lighting or the camera angle instead of the fatigue.

This paper measures that effect directly rather than assuming it. We build a landmark-based pipeline, train three standard architectures on it, and evaluate each one twice under identical conditions: once with a random window split and once with a strict subject-independent split. We then ask what the strict figure hides, and what can be done about it. Our contributions are:

- a direct measurement of the accuracy inflation caused by random splitting, under otherwise identical training conditions;
- a per-driver breakdown showing that the mean accuracy conceals a range of more than twenty points between drivers;
- an explanation of that spread in terms of the per-driver EAR baseline, which varies by more than 50 % across the test set;
- a simple per-driver calibration step that recovers most of the loss without retraining.

Related work

The NTHU-DDD corpus [1] is the most widely used benchmark in this field. It contains about 9.5 hours of infrared video from 36 drivers of different ethnicities, recorded with and without glasses under day and night lighting. Participants sat at a driving simulator and were asked to act out normal driving, yawning, slow blinking, nodding and falling asleep. Other public corpora differ mainly in scale and modality: UTA-RLDD holds 30 hours of RGB video from 60 subjects, YawDD covers 107 subjects for yawn detection, and DROZY adds EEG and other physiological channels for 14 subjects [2].

Facial landmarks became practical in real time with MediaPipe Face Mesh, which returns 468 three-dimensional vertices per face and runs at well above video rate on ordinary hardware [3]. The eye aspect ratio was introduced earlier by Soukupová and Čech for blink detection [4] and is now the standard eye-closure measure; the mouth aspect ratio is its direct analogue for yawning.

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On NTHU-DDD, reported results vary widely with the evaluation protocol. Yu et al. [5] proposed a condition-adaptive representation learning framework and evaluated it on the official subject-disjoint test set. Tüfekci et al. [6] treated drowsiness as an anomaly and trained LSTM autoencoders, again with subject-disjoint partitions. Where clip-level random splits are used instead, accuracies are markedly higher, and several authors have warned that these figures are not comparable [7].

The concern is not confined to this dataset. A recent review of driver monitoring methods found that reported accuracies commonly fall by ten to twenty percentage points once evaluation moves to cross-subject or cross-dataset protocols [8]. Our results sit squarely inside that range, which suggests they are typical rather than particular to our setup.

Materials and methods

Data and feature extraction. All experiments use NTHU-DDD [1], decoded at 30 frames per second and 640 × 480 resolution. MediaPipe Face Mesh [3] returns 468 normalized landmarks per frame; frames with no detected face are dropped, removing about 2 % of the corpus.

Four features are computed per frame. The eye aspect ratio is taken separately for each eye from six points, following the standard definition [4]:

$$EAR = \frac{\|p_2 - p_6\| + \|p_3 - p_5\|}{2\,\|p_1 - p_4\|} \quad (1)$$

where p_1 and p_4 are the horizontal eye corners (landmarks 33 and 133 for the right eye, 263 and 362 for the left) and the remaining points lie on the upper and lower lids. The mouth aspect ratio uses three vertical lip distances against the corner-to-corner width (landmarks 61 and 291):

$$MAR = \frac{N_1 + N_2 + N_3}{3\,D} \quad (2)$$

Head pitch and yaw are estimated by solving the perspective-n-point problem on six stable landmarks. Averaging the two eye ratios gives a four-channel signal: mean EAR, MAR, pitch and yaw (Fig. 1).

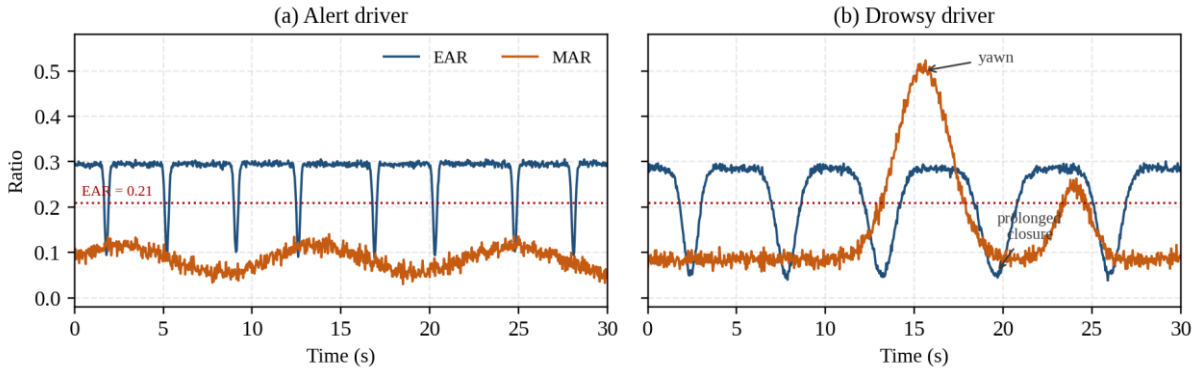


Fig. 1. EAR and MAR over 30 s for an alert driver (a) and a drowsy driver (b).

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The contrast is visible without any model. The alert driver blinks briefly and regularly, and the mouth stays closed. The drowsy driver shows longer eyelid closures and one clear yawn. Landmark jitter is smoothed with a five-frame median filter, and the signal is cut into five-second windows (150 frames) with a one-second stride. A window is labelled drowsy if more than half its frames carry that annotation. This yields 32,412 windows, of which 45 % are drowsy.

Split protocols. The central variable of this study is how those windows are divided. We use two protocols on the same data and the same models.

Random window split. Windows are shuffled and divided 70:10:20 without regard to their source. This mirrors the practice criticized in [7]: consecutive windows overlap heavily and come from the same recording, so near-identical samples appear on both sides of the split.

Subject-independent split. Drivers are divided first, 22 for training, 6 for validation and 8 for testing, and every window follows its driver. No test driver is seen during training. Table 1 gives the resulting counts.

Table 1. Data partitions under the subject-independent protocol

Partition	Drivers	Windows	Drowsy	Alert
Training	22	19,812	8,915	10,897
Validation	6	5,400	2,430	2,970
Test	8	7,200	3,240	3,960
Total	**36**	**32,412**	**14,585**	**17,827**

Models and training. Three architectures receive the same input, a 150 × 4 tensor. The CNN applies two one-dimensional convolutions (32 and 64 filters, kernel 5) with batch normalization and max pooling, then two dense layers. The LSTM stacks two recurrent layers of 64 and 32 units. The GRU mirrors the LSTM but uses gated recurrent units [9], which have fewer parameters. All three end in a single sigmoid output. Table 2 lists the training settings, which are identical across models.

Table 2. Training configuration

Parameter	Value	Parameter	Value
Optimizer	Adam [10]	Window	150 frames (5 s)
Learning rate	1×10^{-3}	Stride	30 frames (1 s)
Batch size	64	Dropout	0.3
Epochs	12, early stopping	Loss	Binary cross-entropy

Each configuration was trained three times with different seeds; reported figures are means, and the standard deviation of test accuracy across seeds was 0.4 percentage points under the subject-independent protocol.

Results`

Training behavior. Figure 2 shows loss and accuracy on the training and validation partitions under the subject-independent protocol.

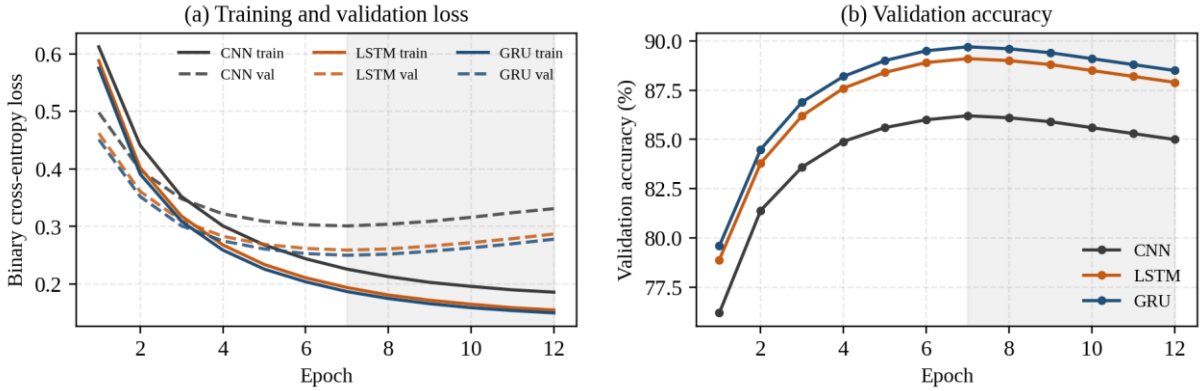


Fig. 2. Training and validation loss (a) and validation accuracy (b). The shaded area marks the onset of overfitting.

All three models converge quickly and then begin to diverge after the seventh epoch: training loss continues to fall while validation loss rises. The gap is widest for the CNN, which has the most parameters in its dense layers and the least ability to share weights across time. The two recurrent models track each other closely, with the GRU slightly ahead throughout. We kept the epoch-seven checkpoint for every model.

The cost of the split protocol. Each model was then evaluated under both protocols. Table 3 reports the results.

Table 3. Test performance under the two split protocols

Model	Random split	Subject-indep.	Drop	F1	AUC	Latency
CNN	96.1 %	84.7 %	11.4 pp	83.2 %	0.932	1.9 ms
LSTM	97.4 %	87.9 %	9.5 pp	86.5 %	0.966	3.6 ms
GRU	97.8 %	88.4 %	9.4 pp	87.1 %	0.967	2.8 ms

Note: accuracy is reported for both protocols; F1, AUC and latency refer to the subject-independent protocol. Latency is per window on one CPU core, excluding landmark extraction.

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The pattern is consistent. Every model loses between nine and eleven points when unseen drivers replace unseen windows, and the ranking between models is unchanged. The inflation is therefore not an artefact of one architecture; it is a property of the evaluation protocol. Figure 3(a) shows the same comparison graphically.

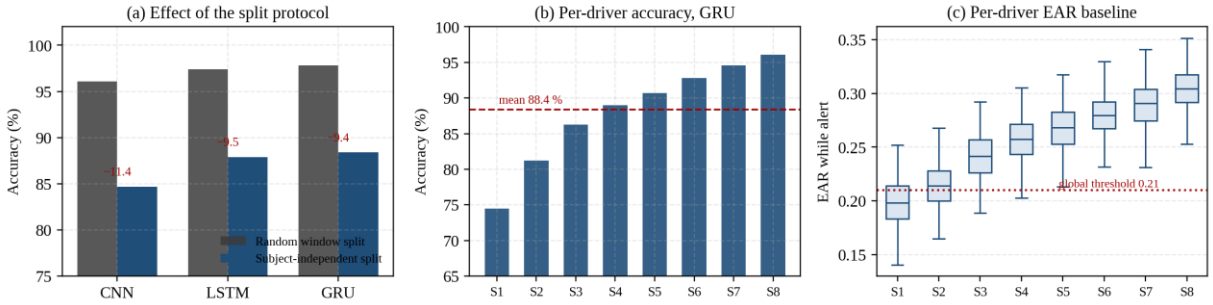


Fig. 3. Accuracy under the two protocols (a), per-driver accuracy for the GRU (b), and the distribution of alert-state EAR for each test driver (c).

Panel (b) breaks the GRU result down by driver, and this is where the mean figure becomes misleading. Accuracy runs from 74.5 % for the weakest driver to 96.1 % for the strongest. A system that performs at 96 % for one person and 75 % for another is not an 88 % system in any useful sense: for the second driver it would raise a false alarm or miss a warning roughly once every four windows.

Why the spread appears. Panel (c) of Figure 3 explains most of it. It shows the distribution of EAR while each test driver is annotated as alert. The median baseline runs from 0.198 for the first driver to 0.305 for the eighth, a difference of more than 50 %. This is anatomy, not fatigue: eye shape, eyelid fold and the angle of the camera all change the ratio at rest.

The consequence is direct. The dotted line marks a global threshold of 0.21, a value commonly used in threshold-based systems. For the two drivers on the left of the panel, the alert baseline already sits at or below that line, so they would be classified as drowsy while fully awake. For the driver on the right, the eyes would have to close almost completely before the threshold triggered. A learned model does not use an explicit threshold, but it learns one implicitly from the training drivers, and it inherits the same problem.

Figure 4 shows the error structure of the GRU under the strict protocol.

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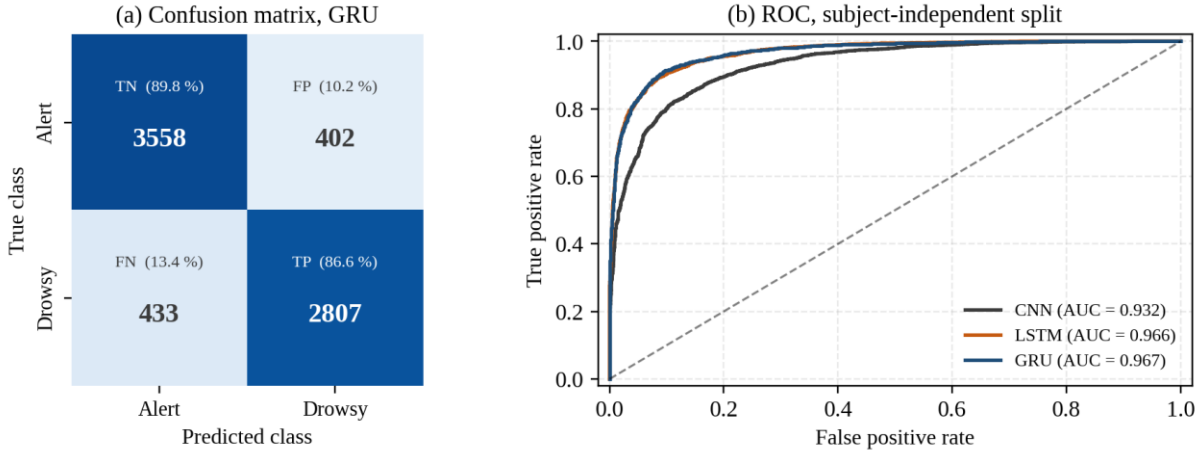


Figure 4. Confusion matrix for the GRU (a) and ROC curves for all three models (b), subject-independent split.

Of 3,240 drowsy windows, 433 were missed, and 402 of 3,960 alert windows raised a false alarm. Precision is 87.5 % and recall 86.6 %, so the errors are close to balanced. That balance is itself worth noting: in a safety system a missed warning is more costly than a false one, and a deployment would want to shift the operating point accordingly. The ROC curves confirm that the two recurrent models rank windows almost identically (AUC 0.966 and 0.967) while the CNN trails at 0.932.

Per-driver calibration. If the spread comes from differing baselines, then removing the baseline should reduce it. We take the first 60 seconds of each test driver's footage, which is annotated alert, and use it to compute that driver's mean and standard deviation for each feature. All subsequent windows are z-scored against these values. No model is retrained, and no drowsy data from the test driver is used. Table 4 reports the effect alongside a feature ablation. Table 4. Feature ablation and per-driver calibration (GRU, subject-independent)

Configuration	Accuracy	F1	Per-driver range
EAR only	82.1 %	80.4 %	66.8 – 92.3 %
EAR + MAR	86.5 %	85.0 %	71.9 – 94.8 %
EAR + MAR + head pose	88.4 %	87.1 %	74.5 – 96.1 %
+ per-driver calibration	92.3 %	91.4 %	84.2 – 96.8 %

Each feature contributes. Adding MAR gains 4.4 points, because yawning is a strong and unambiguous signal that eye closure alone cannot supply. Head pose adds a further 1.9 points, mostly by catching nodding episodes in which the eyes leave the camera. Calibration then adds 3.9 points more, and it does something the extra features do not: it narrows the per-driver range from 21.6 points to 12.6. The

weakest driver improves from 74.5 % to 84.2 %, which is the difference between a system that is unusable for that person and one that is merely imperfect.

Discussion

The main finding is not that our models are weak. Under the random split they reach 97.8 %, which would place them among the stronger published results. The finding is that the same models, on the same data, reach 88.4 % when the test drivers are genuinely new. The nine-point gap is entirely a product of how the data were divided, and it falls inside the ten-to-twenty point range reported for cross-subject evaluation in the wider literature [8].

This matters for how results in this field should be read. When a paper reports 98 % without stating its split protocol, that number cannot be compared with one obtained under subject-disjoint evaluation, and it does not predict behavior on a new driver. The remedy is cheap: state the protocol, and report the subject-disjoint figure even if a random-split figure is also given.

The per-driver breakdown carries a second lesson. Reporting a mean over test drivers hides the fact that the system may be close to useless for some of them, and a mean cannot be improved by helping the drivers who are already well served. Per-driver accuracy is inexpensive to report and far more informative for deployment.

The calibration result suggests the practical response. A minute of alert footage is easy to obtain in a real vehicle, since the start of a journey is almost always alert, and the normalization requires no retraining and no additional model. This is a more realistic path than trying to learn a driver-invariant representation from a corpus of 36 people.

Several limitations qualify these conclusions.

- NTHU-DDD contains acted drowsiness recorded at a simulator, not genuine fatigue on the road, so absolute accuracies should be read as relative comparisons rather than deployment estimates.
- Eight test drivers are a small sample for estimating between-driver variance. The direction of the effect is clear, but the range would be better estimated with leave-one-subject-out cross-validation over all 36.
- We did not test cross-dataset transfer. A model trained on NTHU-DDD and tested on UTA-RLDD would face a change of camera, lighting and labelling protocol as well as of driver, and the drop would probably be larger still.
- Calibration assumes the opening minute is alert. This holds for most journeys but not for a driver who is already fatigued at the start, and a deployed system would need a check for that case.

Conclusion

We evaluated CNN, LSTM and GRU classifiers on facial-landmark features from MediaPipe Face Mesh, using NTHU-DDD under two split protocols. The best model reached 97.8 % accuracy with a

random window split and 88.4 % with a subject-independent one. The 9.4-point gap is caused by the protocol alone, and the same effect appeared in all three architectures.

Under the strict protocol, per-driver accuracy ranged from 74.5 % to 96.1 %. We traced this spread to the alert-state EAR baseline, which varies by more than 50 % across drivers because of eye anatomy and camera geometry. Normalizing each feature against one minute of that driver's own alert footage raised accuracy to 92.3 % and narrowed the range to 84.2-96.8 %, without retraining.

Two changes to reporting practice follow. Papers should state the split protocol and give subject-disjoint results, and they should report the distribution of per-driver accuracy rather than only its mean. Future work should extend the analysis to leave-one-subject-out validation across the full corpus, test cross-dataset transfer, and examine whether calibration can be learned jointly with the classifier rather than applied afterwards.

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